10. A* Algorithm

Aim:

To write a Python program to implement the A* search algorithm for finding the shortest path.

Algorithm:

1. Start the program.
2. Define the graph and heuristic values.
3. Initialize the starting node.
4. Calculate the cost using the formula:

\(f(n)=g(n)+h(n)\)

5. Select the node with the lowest f-value.
6. Expand the selected node.
7. Update the costs of neighboring nodes.
8. Repeat until the goal node is reached.
9. Display the shortest path and total cost.
10. Stop.

INPUT

OUTPUT

```
Path: ['A', 'B', 'D', 'G']
Cost: 6
```

Result:

Thus, the Python program to implement the A* algorithm was executed successfully.